CSEC PHYSICS

May/Jun 2022 – Paper 2

Solutions

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SECTION A

Answer ALL questions.

1. The activity of a radioactive substance was measured over a six minute period. The results are recorded in Table 1.

TABLE 1: ACTIVITY, A AND CORRESPONDING TIME, t

Time, t in minutes	Activity, A Disintegrations per minute
0	80.0
1	50.0
2	35.0
3	22.0
4	13.0
5	8.0
6	4.0

(a) Using the grid provided on Figure 1 on page 5, plot a graph of activity, A versus time, t, and draw a smooth curve. (8 marks)

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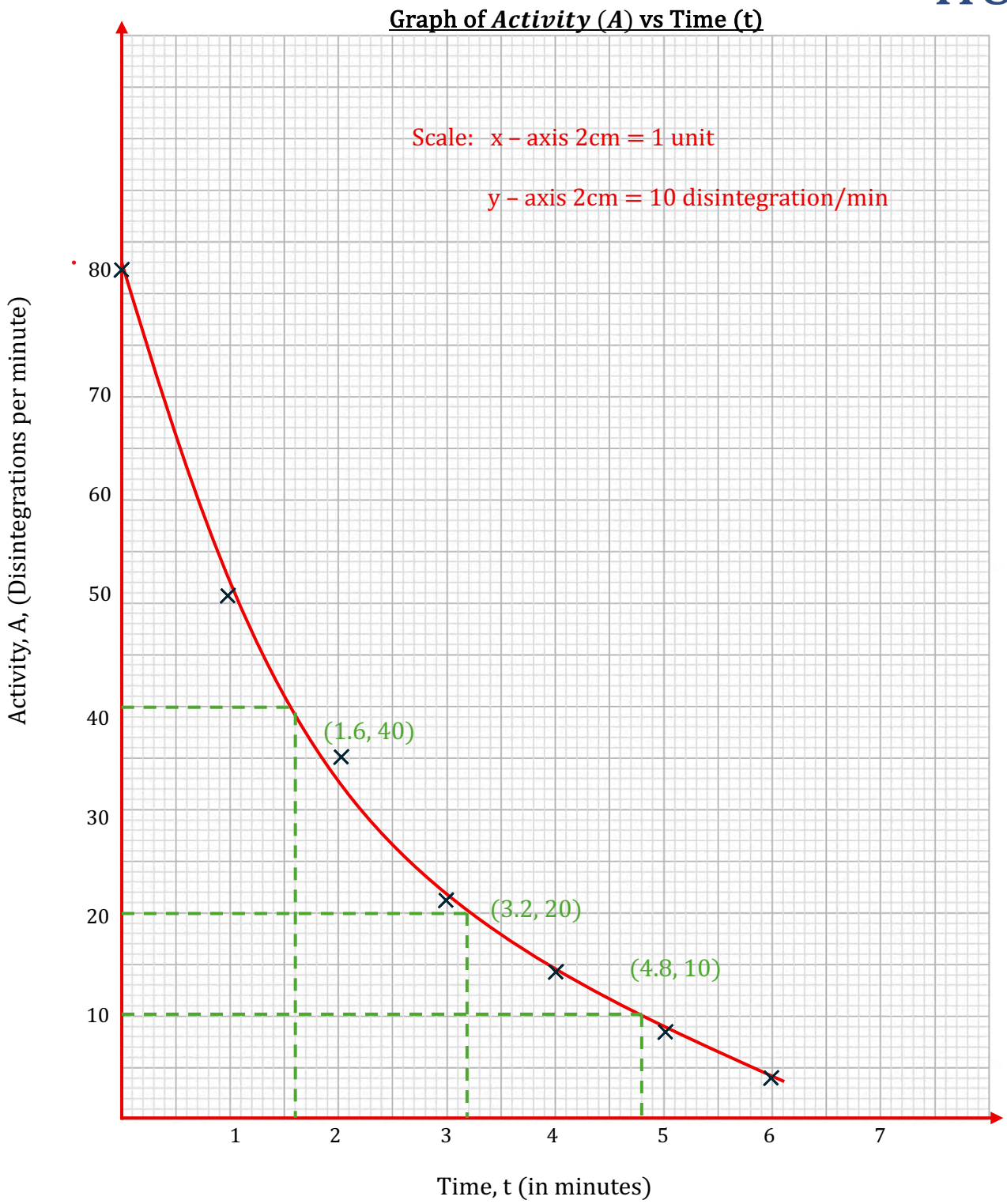

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(b) (i) Define the term half life. (2 marks)

Half-life is the time taken for the number of radioactive nuclei in a sample to halve or the time taken for the number of decays, or activity to halve.

(ii) From your graph on page 5, calculate the average half life of a radioactive substance. (5 marks)

Calculation 1:	Calculation 2:
When a=80, t=0mins	When a=40, t=1.6mins
Half life occurs at $\frac{80}{2} = 40$	Half life occurs at $\frac{40}{2} = 20$
When a=40, t = 1.6 (from graph)	When a=20, t = 3.2 (from graph)
Half life = 1.6 - 0	Half life = 3.2 - 1.6
=1.6 minutes	=1.6 minutes

Average Half life = 1.6 minutes

(iii) On your graph, use dotted lines to determine how long it would take for the activity of the sample to be reduced to ten disintegrations per minute from its original activity level.

Time taken: 4.8 minutes (2 marks)

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(c) (i) Define the term 'nuclear fission' (1 mark)

Nuclear fission is the splitting of large unstable nuclei into smaller, more stable nuclei.

(ii) The equation below is an example of a nuclear fission reaction.

$$^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{90}_{36}Kr + {}^{144}_{56}Ba + 2{}^{1}_{0}n + energy$$

Using the atomic mass data for the nuclides given in table 2, calculate the energy released in the reaction. State your answer in joules.

TABLE 2: NUCLIDE AND CORRESPONDING ATOMIC MASS

Nuclide	Atomic Mass/u
$^{235}_{92}U$	235.118
$^{144}_{56}Ba$	143.881
$^{90}_{36}Kr$	89.847
$^{1}_{0}n$	1.009

[where $u = 1.66 \times 10^{-27} kg, c = 3.0 \times 10^{8} ms^{-1}$]

Mass of Reactants:

$^{235}_{92}U + {}^{1}_{0}n = 235.118 + 1.009$

$= 236.127u$

Mass of Products:

$^{90}_{36}Kr + {}^{144}_{56}Ba + 2{}^{1}_{0}n = 89.847 + 143.881 + 2(1.009)$

$= 235.746u$

Mass Defect (Δm) $= 236.127 - 235.746$

$= 0.381u$

$= (0.381 \times 1.66 \times 10^{-27})$

$= 6.3246 \times 10^{-28} kg$

$\Delta E = \Delta mc^2$

$= (6.3246 \times 10^{-28})(3 \times 10^8)^2$

$= 5.69 \times 10^{-11} J \ to \ 3 \ sf$

(7 marks)

2. (a) Complete Table 3 by inserting the missing physical quantities and their derived SI units.

TABLE 3: PHYSICAL QUANTITIES AND DERIVED SI UNITS

Physical Quantity	**Derived SI Unit**
Area	m^2
Density	kgm^{-3}
Force	$kgms^{-2}$
Velocity	ms^{-1}

(4 marks)

(b) State Newton's second law of motion. (3 marks)

The force applied to a body is directly proportional to its mass and its resultant acceleration. It is given by the equation F=ma

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(c) (i) A basketball has a mass of 500g. Calculate its weight in newtons.

[Acceleration due to gravity, $g = 10ms^{-2}$]

m = 500g

= 0.5kg

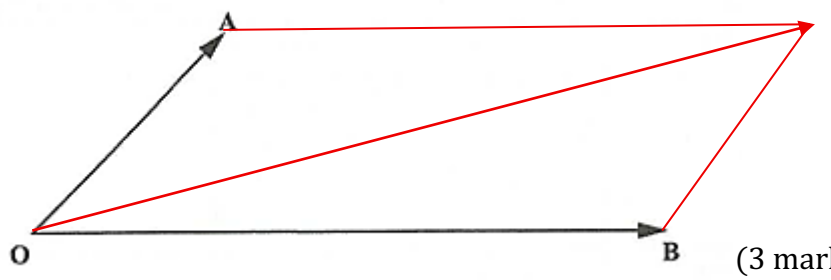
a $= 10ms^{-2}$

Therefore W = mg

=(0.5)(10)

=5N

(ii) Figure 2 is a scale drawing of a vector diagram representing a basketball's velocity, OA, and the wind's velocity OB. Complete the scale drawing on Figure 2 to determine the resultant velocity of the basketball.



(3 marks)

(iii) Determine the magnitude of the resultant velocity in ms^{-1} and its direction in degrees from the wind's velocity, OB. (2 marks)

Measured Diagonal = 8.4 units (to be measured on actual exam paper)

Velocity = 8.4 units (12^0 Anticlockwise from the horizontal as shown in the diagram.

Total 15 marks

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3. (a) Complete Table 4 by inserting the types of thermometers, their use and range.

TABLE 4: TYPE OF THERMOMETER, USE AND RANGE

Type of Thermometer	Use	Range
Thermocouple	Measures extremely high, rapidly changing temperatures	-200^{0}C to 1500^{0}C
Clinical thermometer	Medical- to take human body temperatures	34^{0}C to 43^{0}C
Liquid in glass thermometer Mercury thermometer Laboratory thermometer	Mainly measures boiling and freezing point of water, and room temperature	-37^{0}C to 356^{0}C

(5 marks)

(b) State the value of the ice point on the

(i) Celsius Scale: _____0^{0}C _____

(ii) Kelvin Scale: _____273K_____

(2 marks)

Total 15 marks

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(c) The graph in Figure 3 shows how the pressure, p, varies with the inverse volume, 1/v.

(i) State the law associated with the graph in Figure 3. (1 mark)

Boyle's Law $(P = \frac{K}{V})$

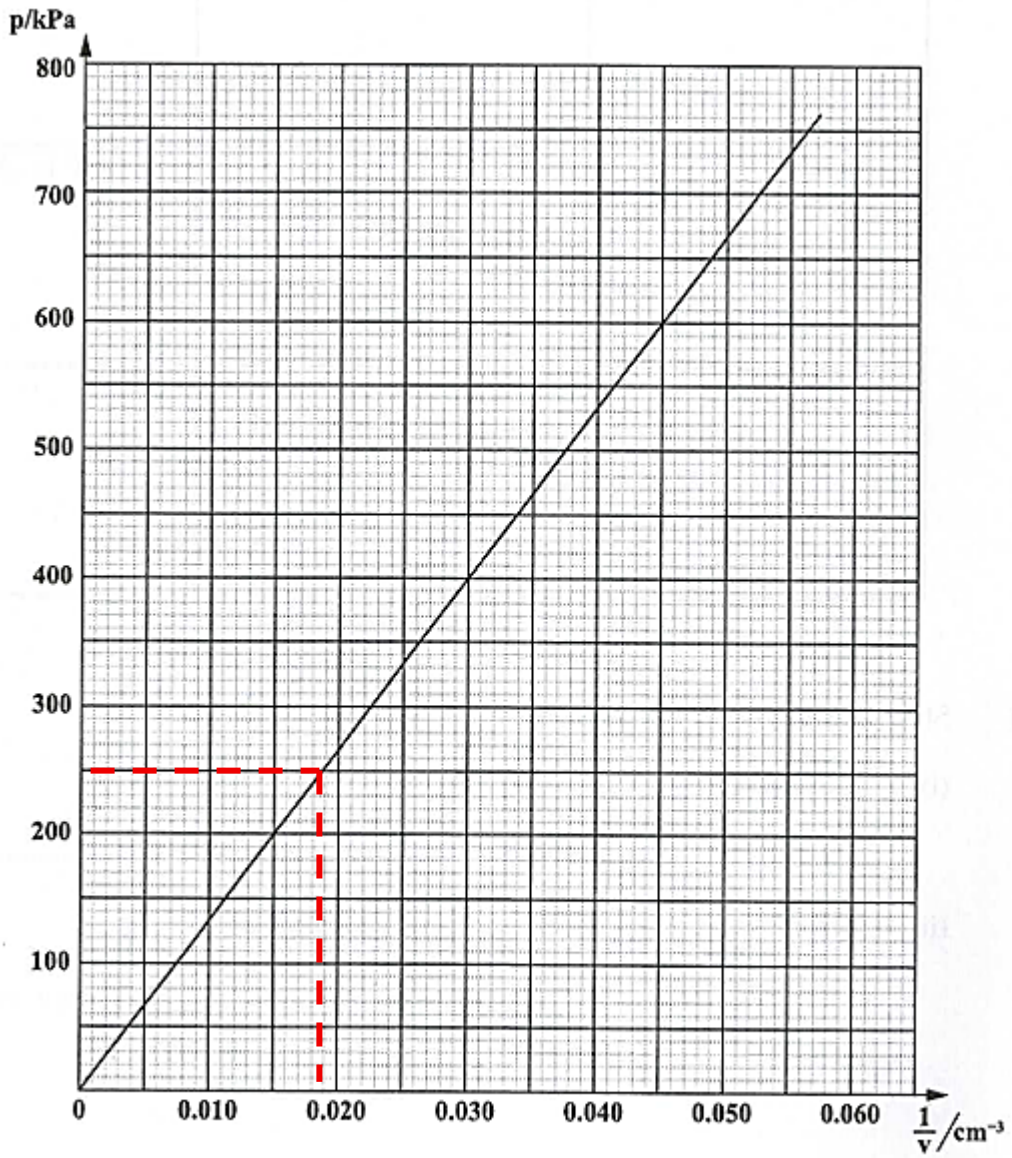

Figure 3. Graph of pressure, *p*, versus inverse volume, *1/v*

(ii) Use the graph in Figure 3, on page 12, to determine the volume when the pressure is 250 kPa. (3 marks)

From Graph,

When P =250kPa, $\frac{1}{v} = 0.019cm^{-3}$

$$\frac{1}{v} = 0.019cm^{-3}$$

$$v = 1/.019$$

$$= 52.6\ cm^3$$

(iii) Use your answer in (c)(ii) to calculate the new volume when the pressure is increased from 250kPa to 975 kPa. (4 marks)

$$P_1V_1 = P_2V_2$$

$$(250)(52.6) = (975)V_2$$

$$V_2 = \frac{13150}{975}$$

$$= 13.5cm^3$$

Total 15 marks

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SECTION B

Answer ALL questions.

4. (a) (i) Define the term 'pressure'. State its SI unit. (3 marks)

Pressure is the normal force acting on a surface per unit area, given by the equation $P = \frac{F}{A}$ units Pa

(ii) Calculate the liquid pressure that a diver experiences at a depth of 0.5km below the surface of the sea, given that acceleration due to gravity, $g = 10Nkg^{-1}$ and the density of the sea water, $\rho = 1025kgm^{-3}$.

$\underline{\rho = 1025kgm^{-3} \qquad g = 10Nkg \qquad h = 0.5km = 500m}$

Pressure $= \rho gh$

$=(1025)(10)(500)$

$=5.125 \times 10^6$ Pa

(4 marks)

(b) (i) State Archimedes' principle. (3 marks)

The upwards force that acts on an object wholly or partially immersed in a fluid is equal and opposite to the weight of the fluid displaced by the object.

Upthrust = Weight of the fluid displaced

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(ii) The boat which has a weight of 83 000N floats in sea water. If the density of the sea water is $1025kgm^{-3}$, calculate the volume of the sea water displaced by the boat. (5 marks)

$Weight = 83000N$ $\rho_{seawater} = 1025kgm^{-3}$

According to Archimedes Principle

$Weight\ of\ Water\ Displaced = Weight\ of\ Boat$

$= 83000N$

$W = mg$

$m = \frac{W}{g}$

$m = \frac{83000}{10}$

$m = 8300kg$

Therefore

$\rho = \frac{m}{v}$

$v = \frac{m}{\rho}$

$v = \frac{8300}{1025}$

$v = 8.10m^3$

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5. (a) Define the term 'longitudinal wave (2 marks)

A longitudinal wave is a type of mechanical wave in which particles of the medium vibrate parallel to the direction of the wave propagation, creating alternating regions of compression and rarefaction.

(b) Figure 4 is a wave train on which the points K to S and a to d are shown.

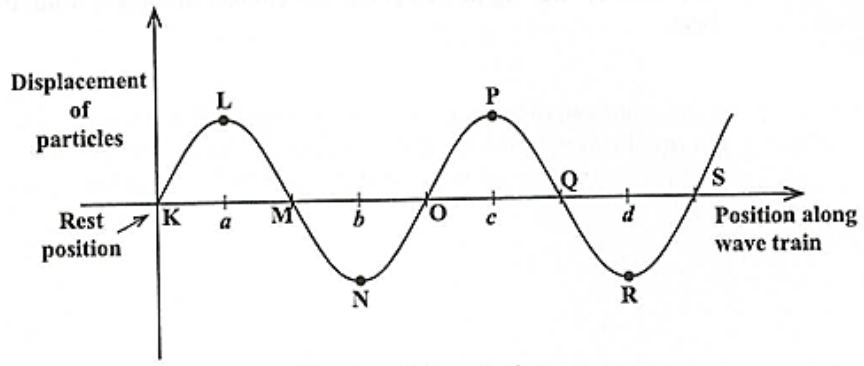


Figure 4. Wave train

Write TWO letters whose distance between them on the wave train in Figure4 represents (2 marks)

(i) the amplitude: _____ L, a _____

(ii) A wavelength: _____L, P or M, Q _____

(c) State TWO differences between 'light waves' and 'sound waves'

Light	Sound
Transverse wave	Longitudinal wave
Can travel in a vacuum	Needs a medium
Speed = c ($3 \times 10^8 ms^{-1}$)	Speed = $330 ms^{-1}$

(2 marks)

(d) An object is placed 15cm in front of a converging lens with a focal length 10cm.

(i) Calculate the image distance, v. (4 marks)

Lens Formula: $\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$

$$\frac{1}{10} = \frac{1}{15} + \frac{1}{v}$$

$$\frac{1}{v} = \frac{1}{10} - \frac{1}{15}$$

$$\frac{1}{v} = \frac{1}{30}$$

$$v = 30 \text{ cm}$$

(ii) Determine the magnification of the image formed. (2 marks)

$$Mag = \frac{Image\ distance}{Object\ distance}$$

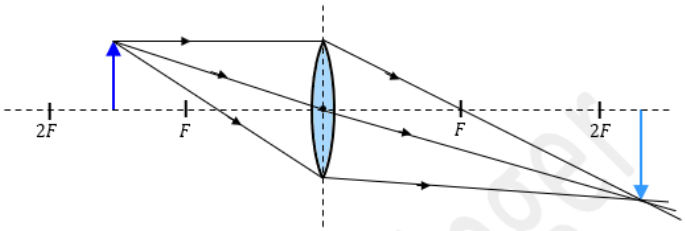
$$= \frac{30}{15}$$

$$= 2$$

(iii) State the nature and position of the image formed. (3 marks)

The image will be found beyond 2F on the opposite side of the lens.

Image will be inverted and magnified.



Total 15 marks

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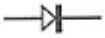
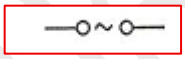
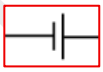
6. (a) Complete Table 5 by inserting the missing electrical circuit symbols and their names.

TABLE 5: ELECTRICAL CIRCUIT SYMBOLS AND NAMES

Name	Circuit Symbol
Filament lamp	
Diode	
Fuse	
AC Power Supply	
Cell	
Rheostat/ Variable Resistor	

(6 marks)

(b) The manufacturer of a popular cellphone has upgraded the battery capacity to 9600C.

(i) If the standard charger can deliver a current of 0.8A, calculate the time taken to charge the battery. (3 marks)

$Q = 9600C \qquad I = 0.8A$

$Q = It$

$t = \frac{Q}{I}$

$t = \frac{9600}{0.8}$

$t = 12000s$

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(ii) Calculate the voltage, given that the power supply of the cellphone is 4.4W. (3 marks)

$P = IV$

$V = \frac{P}{I}$

$V = \frac{4.4}{0.8}$

$V = 5.5V$

(iii) Determine the work done to fully charge the cellphone battery.

$E = Pt$

$= (4.4)(12000)$

$= 52800J$

(3 marks)

Total 15 marks

END OF TEST

IF YOU FINISH BEFORE TIME IS CALLED, CHECK YOUR WORK ON THIS TEST.

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